

Ahmad Fathan Hidayatullah, PhD



Assistant Professor in Computer Science with expertise in text mining, natural language processing, data science, and artificial intelligence. Actively engaged in academic collaboration, with a strong commitment to impactful research and scholarly excellence.



Contact

Office: Department of Informatics, Faculty of Industrial Technology, Universitas Islam Indonesia

Office Address: Jl. Kaliurang Km. 14.5, Yogyakarta 55584, Indonesia

Phone: (0274) 898 444

Home: Jonggrangan RT 003/RW 018, Sumberadi, Mlati, Sleman, Yogyakarta 55288

Phone: +62 878 939 463 95

Email: fathan@uii.ac.id | fathanel@gmail.com

Education

- PhD in Digital Science (Computer Science), School of Digital Science, Universiti Brunei Darussalam, August 2021 – 2024 (UBD Graduate Scholarship Awardee)
- Master of Computer Science (M.Cs.), Universitas Gadjah Mada, 2011 - 2014
- Bachelor of Engineering (B.Eng.), Universitas Islam Negeri Sunan Kalijaga Yogyakarta, 2006 - 2010

Professional Experience

- Lecturer, Department of Informatics, Universitas Islam Indonesia, 2014 – Present
- Head of Information System Development, Board of Information System, Universitas Islam Indonesia (2016-2021) and (February 2025-present)
- Researcher, Center of Data Science, Department of Informatics, Universitas Islam Indonesia, 2018 - Present
- Researcher, Informatics Theory and Intelligent Systems Cluster, Universitas Islam Indonesia, 2015 – Present
- Researcher, Information System Cluster, Universitas Islam Indonesia, 2015 - Present

Previous Professional Experience

- Lecturer, STMIK Jenderal Achmad Yani Yogyakarta, 2011 - 2014
- Lecturer, Universitas Teknologi Yogyakarta, 2012 - 2014
- Head of Education and Da'wah Division, Directorate of Islamic Education and Development, December 2014 - March 2016
- Head of Information System Development, Board of Information System, Universitas Islam Indonesia, April 2016 - June 2021

Professional Affiliations

- Member, Indonesia Association of Computational Linguistics (INACL), 2016 – Present
- Member, Indonesia Association of Data Scientists (AIDI), 2019 – Present
- Member of IEEE

Research Interests

- Text Mining
- Natural Language Processing
- Data Science
- Artificial Intelligence

Skills Summary

- University Teaching
- Programming Languages: Python, R
- Over 10 years of research experience in text mining, natural language processing, data science.

Research Grants

- **2020:** Dialect Identification for Javanese Language using Deep Learning
- **2019:** Adult Content Detection from Short Text in Bahasa Indonesia using Long Short-Term Memory (LSTM) Networks
- **2019:** Topic Modeling for Sentiment Analysis on Indonesian Presidential Election using Twitter Data
- **2018:** Discovering Infrastructure Development Topic in Indonesia using LDA Topic Modeling Approach
- **2018:** Topic Modeling of Weather and Climate Condition on Twitter Using Latent Dirichlet Allocation (LDA)

Professional Service

- **Journal Reviewer**
 - Applied Intelligence
 - Engineering Applications of Artificial Intelligence
 - Expert Systems with Applications

- Heliyon
- Intelligence-based medicine
- Informatics in Medicine
- International Journal of Information Management Data Insights
- IAES International Journal of Artificial Intelligence (IJ-AI)
- Social Network Analysis and Mining
- IEEE Access
- Connection Science
- Cybernetics & Systems
- Jordanian Journal of Computers and Information Technology
- PeerJ Computer Science
- Natural Language Processing
- **Speaker**
 - *Text Mining for Cybersecurity*, Research Dissemination and Focus Group Discussion Text Mining, Politeknik Negeri Cilacap, 3 December 2025.
 - *Data Science Fundamental Training*, In House Training Data Science, Universitas Tiga Serangkai, 21 November 2025.
 - *Implementing AI in Education: Optimizing Academic Tasks*. Education Pre Conference Workshop The 5th International Conference on Pharmaceutical Research and Practice (ICPRP) 2024. The Department of Pharmacy, Universitas Islam Indonesia. 13 September 2024.
 - *Integrating Artificial Intelligence and Social Media for Education*. Seminar Nasional, Pameran Alat Peraga, dan Kompetisi Matematika (SEMPOA) IX. The Department of Mathematics Education, Universitas Muhammadiyah Surakarta. 24 June 2024.
 - *Large Language Model for Text Analysis*, Online Lecture Series Program 2024. Directorate Office of International Affairs, Universitas Islam Indonesia. 7 June 2024.
 - *Build Machine Learning Web Apps using Gradio*, Bootcamp Satria Data Universitas Islam Indonesia. The Department of Statistics, Universitas Islam Indonesia. 18 May 2024.
 - *Introduction to Text Mining*. Online Summer Course Vocational College Universitas Gadjah Mada 2023. The Department of Electrical Engineering and Informatics, Universitas Gadjah Mada. 3–14 July 2023.

Publications (2021-2026)

- Shafik, Wasswa, **Ahmad Fathan Hidayatullah**, Kassim Kalinaki, Haji Gul, Rufai Yusuf Zakari, and Ali Tufail. "A systematic literature review on transparency and interpretability of AI models in healthcare: taxonomies, tools, techniques, datasets, open research challenges, and future trends." *Health and Technology* (2026): 1-22.
- **Hidayatullah, Ahmad Fathan**, and Abdullah Yahya Al-Sabahi. "Sentiment Analysis on Indonesian-Javanese-English Code-Mixed Texts via Parameter-Efficient Fine-Tuning and

Selective Layer Unfreezing." In *2025 International Conference on Information and Communication Technology (ICoICT)*, pp. 1-6. IEEE, 2025.

- Aslam, Muhammad Muzamil, Wasswa Shafik, **Ahmad Fathan Hidayatullah**, Kassim Kalinaki, Haji Gul, Rufai Yusuf Zakari, and Ali Tufail. "Intelligent transportation systems: A critical review of integration of cyber-physical systems (CPS) and Industry 4.0." *Digital communications and networks* (2025).
- **Hidayatullah, Ahmad Fathan**, Rosyzie Anna Apong, Daphne Teck Ching Lai, and Atika Qazi. "Pre-trained language model for code-mixed text in Indonesian, Javanese, and English using transformer." *Social Network Analysis and Mining* 15, no. 1 (2025): 1-17.
- Aslam, Muhammad Muzamil, Wasswa Shafik, **Ahmad Fathan Hidayatullah**, Kassim Kalinaki, Haji Gul, Rufai Yusuf Zakari, and Ali Tufail. "Intelligent Transportation Systems: A Critical Review of Integration of Cyber-physical Systems (CPS) and Industry 4.0." *Digital Communications and Networks* (2025).
- **Hidayatullah, Ahmad Fathan**. "Parameter Efficient Fine-tuning using Low-Rank Adaptation for Emotion Classification in Indonesian Texts." In *2024 9th International Conference on Information Technology and Digital Applications (ICITDA)*, pp. 1-6. IEEE, 2024.
- Haryono, Kholid, and **Ahmad Fathan Hidayatullah**. "Large Language Model: Design Mobile Platform for Problem Solving Ideation." In *2024 9th International Conference on Information Technology and Digital Applications (ICITDA)*, pp. 1-7. IEEE, 2024.
- **Hidayatullah, Ahmad Fathan**. "Code-Mixed Sentiment Analysis on Indonesian-Javanese-English Text Using Transformer Models." In *2024 8th International Conference on Information Technology, Information Systems and Electrical Engineering (ICITISEE)*, pp. 340-345. IEEE, 2024.
- **Hidayatullah, Ahmad Fathan**, Rosyzie Anna Apong, Daphne Teck Ching Lai, and Atika Qazi. "Word Level Language Identification in Indonesian-Javanese-English Code-Mixed Text." *Procedia Computer Science* 244 (2024): 105-112.
- **Hidayatullah, Ahmad Fathan**, Kassim Kalinaki, Haji Gul, Rufai Zakari Yusuf, and Wasswa Shafik. "Leveraging Natural Language Processing for Enhanced Text Analysis in Business Intelligence." In *Intersection of AI and Business Intelligence in Data-Driven Decision-Making*, pp. 151-182. IGI Global, 2024.
- Iksan, Nur, Ali Tufail, Rosyzie Anna Apong, and **Ahmad Fathan Hidayatullah**. "Optimizing Maritime Passenger Transfer in Rich Vehicle Routing Problem Using a Hybrid Genetic Algorithm." *IEEE Access* (2024).
- Abubakari, Mussa Saidi, Wasswa Shafik, and **Ahmad Fathan Hidayatullah**. "Evaluating the Potential of Artificial Intelligence in Islamic Religious Education: A SWOT Analysis Overview." In *AI-Enhanced Teaching Methods*, pp. 216-239. IGI Global, 2024.
- Shafik, Wasswa, **Ahmad Fathan Hidayatullah**, Kassim Kalinaki, and Muhammad Muzamil Aslam. "Artificial Intelligence (AI)-Assisted Computer Vision (CV) in Healthcare Systems." In *Computer Vision and AI-Integrated IoT Technologies in the Medical Ecosystem*, pp. 17-36. CRC Press, 2024.

- **Hidayatullah, Ahmad Fathan**, Kassim Kalinaki, Muhammad Muzamil Aslam, Rufai Yusuf Zakari, and Wasswa Shafik. "Fine-Tuning BERT-Based Models for Negative Content Identification on Indonesian Tweets." In *2023 8th International Conference on Information Technology and Digital Applications (ICITDA)*, pp. 1-6. IEEE, 2023.
- **Hidayatullah, Ahmad Fathan**, Rosyzie Anna Apong, Daphne Teck Ching Lai, and Atika Qazi. "Adult Content Detection on Indonesian Tweets by Fine-tuning Transformer-based Models." In *2023 6th International Conference on Applied Computational Intelligence in Information Systems (ACIIS)*, pp. 1-6. IEEE, 2023.
- **Hidayatullah, Ahmad Fathan**, Rosyzie Anna Apong, Daphne TC Lai, and Atika Qazi. "Corpus creation and language identification for code-mixed Indonesian-Japanese-English Tweets." *PeerJ Computer Science* 9 (2023): e1312.
- **Hidayatullah, Ahmad Fathan**, Wahyu Dwi Amirullah, Irving Vitra Paputungan, and Rosyzie Anna Apong. "Closely related Indonesian language identification using deep learning." In *AIP Conference Proceedings*, vol. 2508, no. 1. AIP Publishing, 2023.
- **Hidayatullah, Ahmad Fathan**, Rosyzie Anna Apong, Daphne Teck Ching Lai, and Atika Qazi. "Extracting Tourist Attraction Entities from Text using Conditional Random Fields." In *2022 IEEE 7th International Conference on Information Technology and Digital Applications (ICITDA)*, pp. 1-6. IEEE, 2022.
- Zahra, Annisa, **Ahmad Fathan Hidayatullah**, and Septia Rani. "Bidirectional long-short term memory and conditional random field for tourism named entity recognition." *Int J Artif Intell ISSN 2252*, no. 8938 (2022): 1271.
- **Hidayatullah, Ahmad Fathan**, Atika Qazi, Daphne Teck Ching Lai, and Rosyzie Anna Apong. "A systematic review on language identification of code-mixed text: techniques, data availability, challenges, and framework development." *IEEE access* 10 (2022): 122812-122831.
- Fudholi, DThomas Hatta, Royan Abida N. Nayoan, **Ahmad Fathan Hidayatullah**, and Dede Brahma Arianto. "A Hybrid Cnn-Bilstm Model For Drug Named Entity Recognition." *Journal of Engineering Science and Technology* 17.1 (2022): 0730-0744.
- Cahyaningtyas, Siwi, DThomas Hatta Fudholi, and **Ahmad Fathan Hidayatullah**. "Deep Learning for Aspect-Based Sentiment Analysis on Indonesian Hotels Reviews." *Kinetik: Game Technology, Information System, Computer Network, Computing, Electronics, and Control* (2021).
- Nayoan, Royan Abida N., **Ahmad Fathan Hidayatullah**, and DThomas Hatta Fudholi. "Convolutional Neural Networks for Indonesian Aspect-Based Sentiment Analysis Tourism Review." *2021 9th International Conference on Information and Communication Technology (ICoICT)*. IEEE, 2021.
- Cahyaningtyas, Siwi, DThomas Hatta Fudholi, and **Ahmad Fathan Hidayatullah**. "Deep Learning for Aspect-Based Sentiment Analysis on Indonesian Hotels Reviews." *Kinetik: Game Technology, Information System, Computer Network, Computing, Electronics, and Control* (2021).

- Nastiti, Kartika Rizqi, **Ahmad Fathan Hidayatullah**, and Ahmad Rafie Pratama. "Discovering Computer Science Research Topic Trends using Latent Dirichlet Allocation." *Jurnal Online Informatika* 6.1 (2021): 17-24.
- Windiatmoko, Yurio, Ridho Rahmadi, and **Ahmad Fathan Hidayatullah**. "Developing Facebook Chatbot Based on Deep Learning Using RASA Framework for University Enquiries." In IOP Conference Series: Materials Science and Engineering, vol. 1077, no. 1, p. 012060. IOP Publishing, 2021.
- **Hidayatullah, Ahmad Fathan**, Muhammad Rifqi Ma'arif, Muhammad Habibie, and Siti Khomsah. "Indonesia Infrastructure Development Topic Discovery on Online News with Latent Dirichlet Allocation." In IOP Conference Series: Materials Science and Engineering, vol. 1077, no. 1, p. 012012. IOP Publishing, 2021.
- **Hidayatullah, Ahmad Fathan**, Siwi Cahyaningtyas, and Anisa Miladya Hakim. "Sentiment Analysis on Twitter using Neural Network: Indonesian Presidential Election 2019 Dataset." In IOP Conference Series: Materials Science and Engineering, vol. 1077, no. 1, p. 012001. IOP Publishing, 2021.
- Fudholi, Dhomas Hatta, Yurio Windiatmoko, Nurdi Afrianto, Prastyo Eko Susanto, Magfirah Suyuti, **Ahmad Fathan Hidayatullah**, and Ridho Rahmadi. "Image Captioning with Attention for Smart Local Tourism using EfficientNet." In IOP Conference Series: Materials Science and Engineering, vol. 1077, no. 1, p. 012038. IOP Publishing, 2021.